Week 1: Data Exploration & EDA Report

1. Dataset Exploratory Summary

The DePaul Dataset contains information related to student applications and enrolment activities at DePaul University.

Dataset Structure

- Total Records (Rows): 7,543
- Total Variables (Columns): 80
- Data Types:
  - Numerical Variables: 50
  - Categorical Variables: 30

Initial Observations

- The dataset contains student demographic, academic, and application-related information.
- Most applicants are from India.
- All records are associated with DePaul University.
- Several columns contain missing values and may require cleaning before advanced analysis.
- No duplicate records were identified.

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2. Data Dictionary

Variable Name	Data Type	Description
Reference_ID	Integer	Unique student/application identifier
Given_Name	Text	Student first name
Last_Name	Text	Student last name
College	Text	College information
Major	Text	Student major
Degree_Type	Text	Degree type
Country	Text	Applicant country
University	Text	University name

Citizenship	Text	Citizenship information
Status	Text	Application status
Intake	Text	Intake term
City	Text	Applicant city
Region	Text	Applicant region/state
Email_ID	Text	Student email
Application_Source	Text	Source of application
Phone_Number	Text	Applicant contact number

Note: Several variables have unclear naming conventions and require further documentation from the dataset provider.

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3. Data Quality Report

Missing Values

Several variables contain substantial missing values. Examples include:

- Most_Recent_Released_Decision
- Date_of_Birth
- Housing_Contract
- Is_Global_Grad
- Is_Admitted
- SEVIS_ID
- RIT_Email_Created
- Official_University_email_address

Many of these columns contain 100% missing values and may be removed during data cleaning.

Duplicate Records

- Duplicate Rows Found: 0

Data Inconsistencies

- Empty fields present in multiple columns.
- Some variables contain incomplete records.
- Standardization may be required for categorical fields.

Recommended Cleaning Actions

1. Remove columns with excessive missing values.
2. Handle null values appropriately.
3. Standardize categorical values.
4. Validate email and contact information fields.

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4. Exploratory Data Analysis (EDA)

Country Distribution

Top Applicant Countries:

Country	Count
India	4617
United States	764
Ghana	440
Nigeria	430
Pakistan	273

Observation:
India contributes the largest number of applications, representing the majority of records in the dataset.

University Distribution

- DePaul University: 100% of records

Observation:
The dataset is focused entirely on applicants associated with DePaul University.

Application Trends

The dataset contains information regarding:

- Student location
- Academic preferences
- Degree type
- Application status
- Recruitment channels

These variables can be used in future analysis to identify enrollment trends and applicant behavior.

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5. Visualizations Before Data Cleaning

The following visualizations were created before thorough data cleaning was performed to ascertain the level of accuracy and integrity of data before analysis:

Chart 1: Country Distribution Bar Chart

Purpose: To identify countries contributing the highest number of applications.

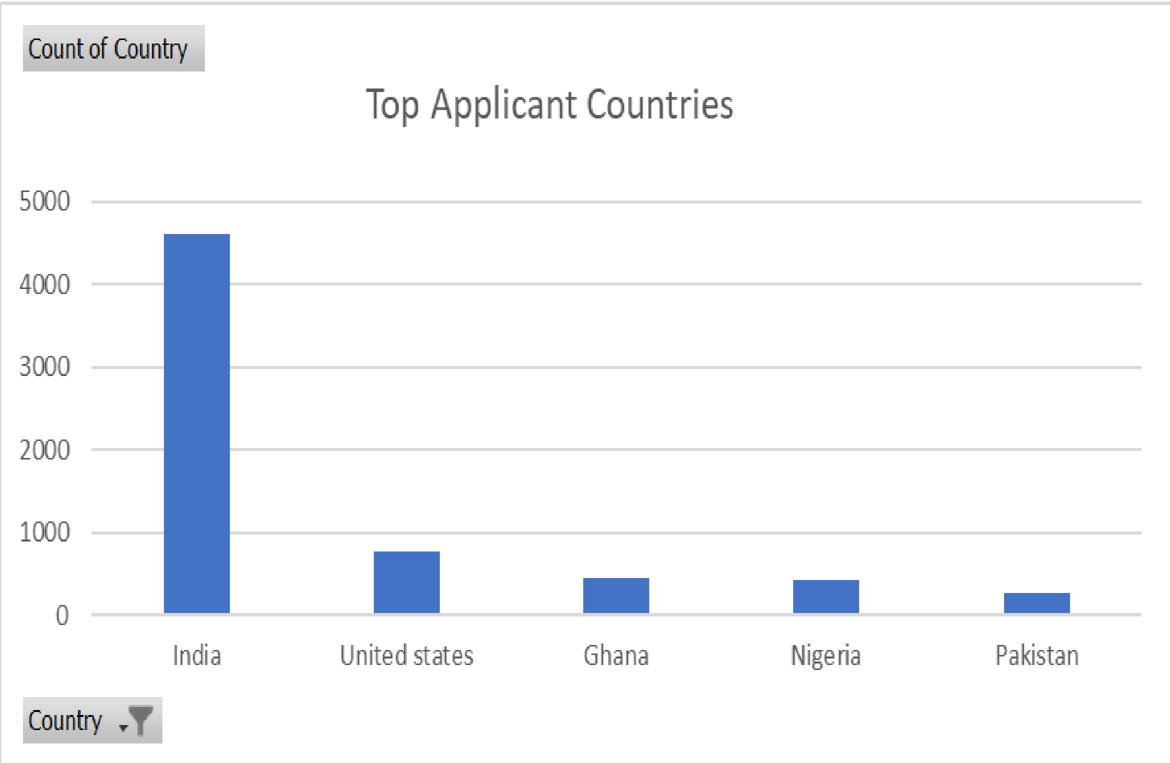


Figure 1: Top Applicant Countries

> The bar chart illustrates the number of applicants from the top five countries. India has the highest number of applicants (4,617), followed by the United States (764), Ghana (440), Nigeria (430), and Pakistan (273). This indicates that the majority of applicants originate from India, highlighting a strong concentration of applications from South Asia.\

Chart 2: Application Status Distribution

Purpose:To understand application progress and outcomes.

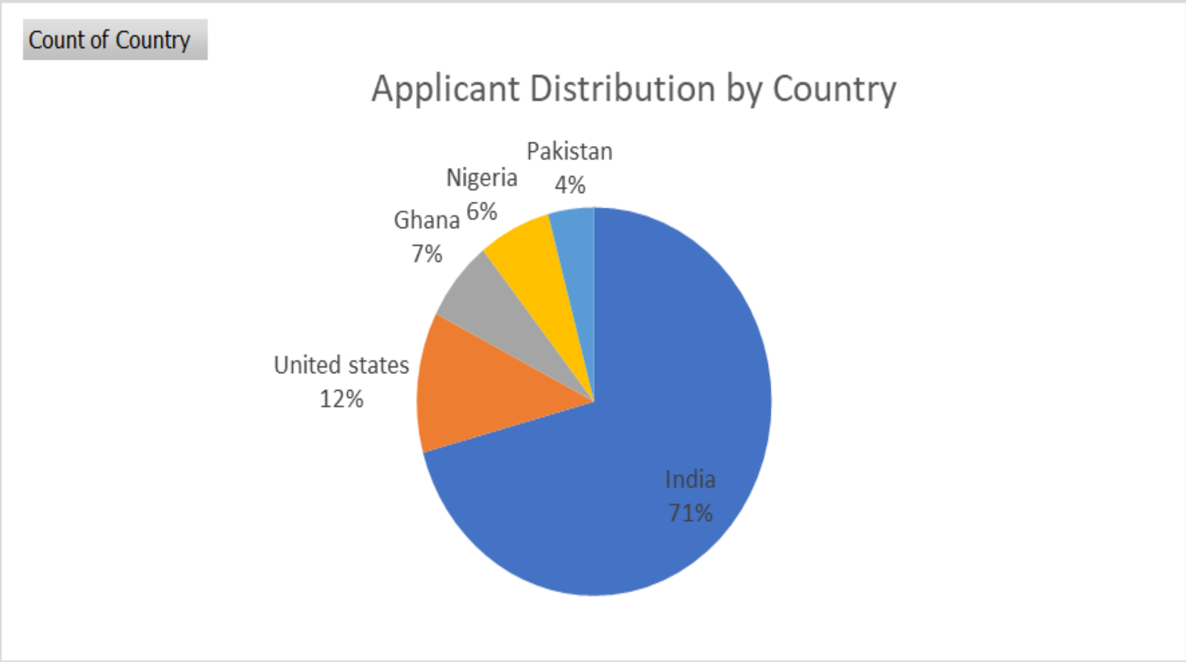


Figure 2: Applicant Distribution by Country

The pie chart shows the percentage distribution of applicants across the top five countries. India represents the largest share of applicants, while the United States, Ghana, Nigeria, and Pakistan contribute smaller proportions.

Chart 3: Intake Distribution

Purpose: To analyse student intake preferences.

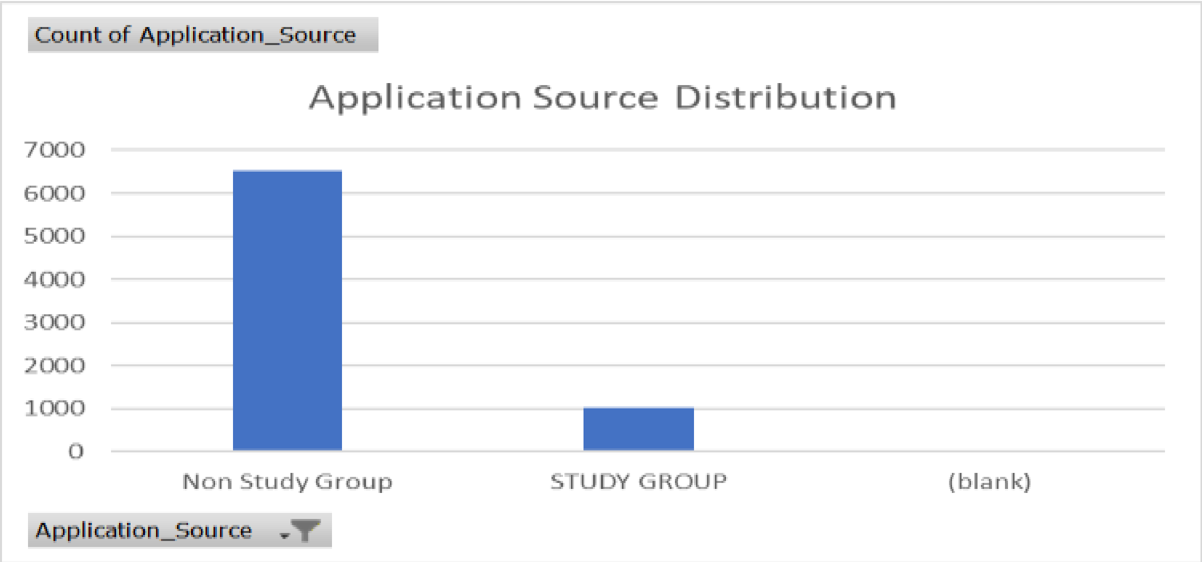
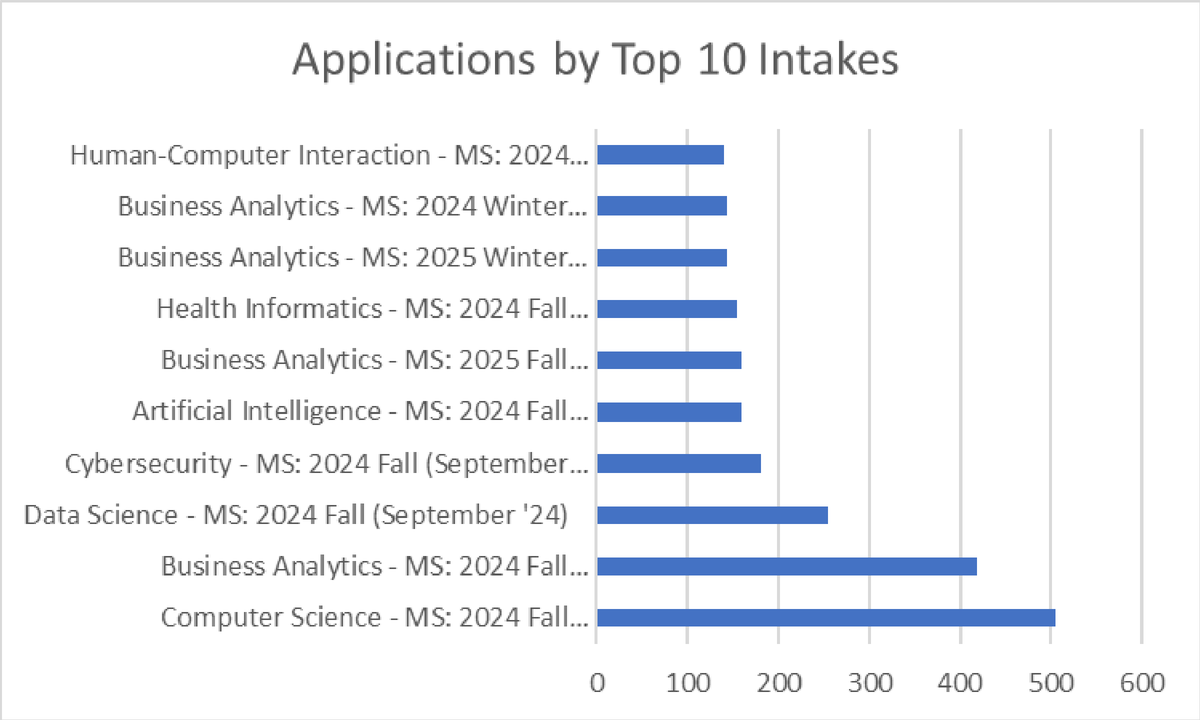


Figure 3: Application Source Distribution

The chart illustrates the distribution of applicants across different application sources. It helps identify the most effective channels through which students apply to DePaul University and provides insights into recruitment performance.

Chart 4: Applications by Top 10 Intakes

Purpose: To identify the most popular degree categories.



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6. Insights Summary

Key Findings

1. The dataset contains 7,543 application records.
2. India is the dominant source country for applicants.
3. No duplicate records were detected.
4. Multiple variables contain significant missing values.
5. The dataset provides extensive information on applicant demographics and academic interests.

Notable Trends

- Strong applicant representation from South Asia.
- Large concentration of records from India.
- University information is highly consistent across records.
- Several administrative fields require cleaning before deeper analysis.

Recommendations for Week 2

- Perform detailed data cleaning.

- Handle missing values systematically.
- Explore relationships between country, degree type, intake, and application status.
- Conduct correlation and trend analysis.
- Build advanced visualizations to support decision-making.

Conclusion

The DePaul Dataset provides a comprehensive view of student application information. Initial exploration identified key demographic trends, data quality concerns, and opportunities for deeper analysis. Further cleaning and advanced analytical techniques will enable more meaningful insights in Week 2.